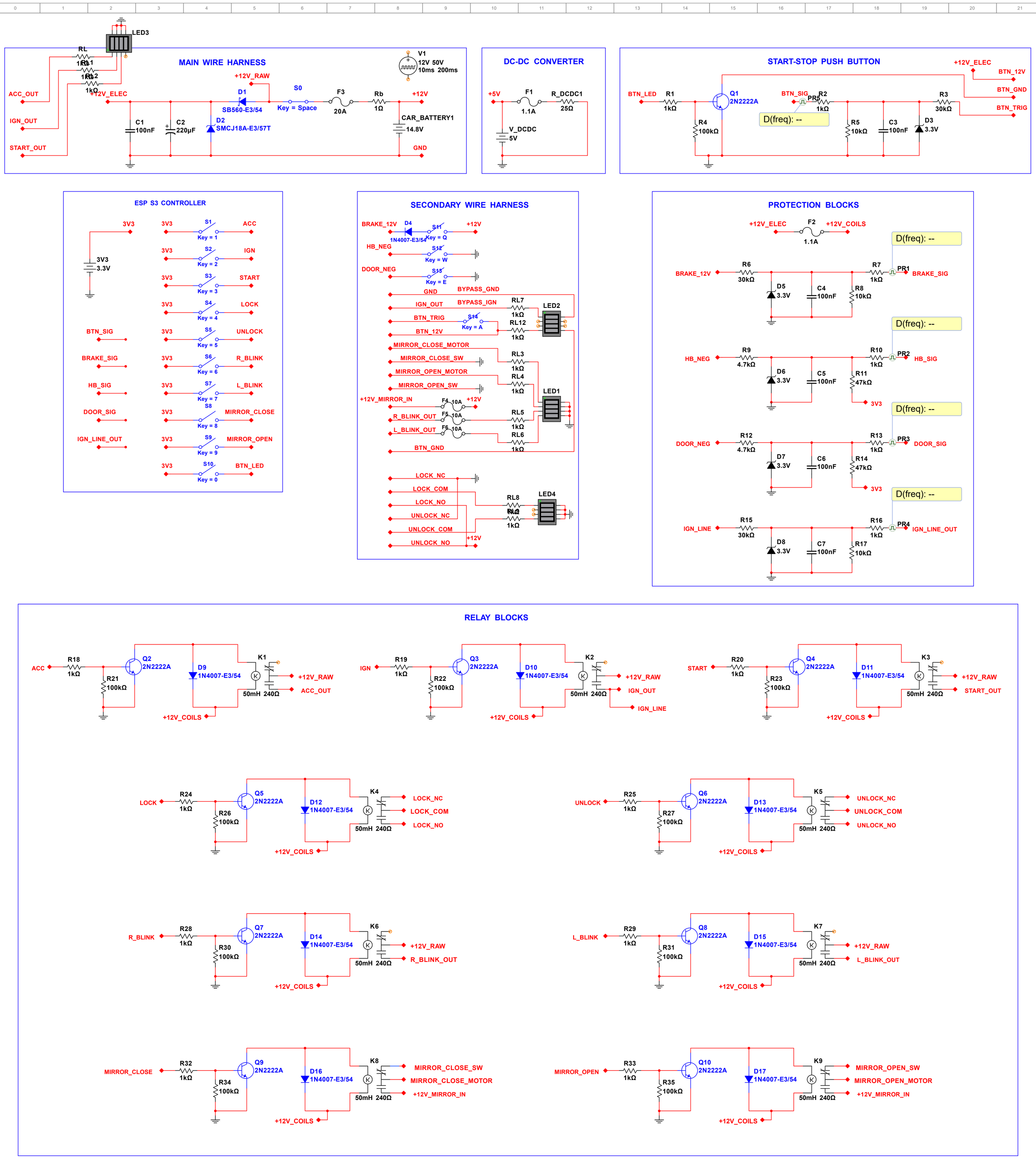
MAIN WIRE HARNESS
LED3
RL
Rb1
Rb2
1kΩ
ACC_OUT
IGN_OUT
START_OUT
+12V_ELEC
C1
100nF
C2
220µF
D1
SB560-E3/54
D2
SMCJ18A-E3/57T
+12V_RAW
S0
Key = Space
F3
20A
Rb
1Ω
+12V
V1
12V 50V
10ms 200ms
CAR_BATTERY1
14.8V
GND
DC-DC CONVERTER
+5V
F1
1.1A
R_DCDC1
25Ω
V_DCDC
5V
START-STOP PUSH BUTTON
BTN_LED
R1
1kΩ
R4
100kΩ
Q1
2N2222A
BTN_SIG
PR5
R2
1kΩ
D(freq): --
R5
10kΩ
C3
100nF
D3
3.3V
R3
30kΩ
+12V_ELEC
BTN_12V
BTN_GND
BTN_TRIG
ESP S3 CONTROLLER
3V3
3.3V
BTN_SIG
BRAKE_SIG
HB_SIG
DOOR_SIG
IGN_LINE_OUT
S1 Key = 1 ACC
S2 Key = 2 IGN
S3 Key = 3 START
S4 Key = 4 LOCK
S5 Key = 5 UNLOCK
S6 Key = 6 R_BLINK
S7 Key = 7 L_BLINK
S8 Key = 8 MIRROR_CLOSE
S9 Key = 9 MIRROR_OPEN
S10 Key = 0 BTN_LED
SECONDARY WIRE HARNESS
BRAKE_12V
D4
1N4007-E3/54
S11 Key = Q
+12V
HB_NEG
S12 Key = W
DOOR_NEG
S13 Key = E
GND
BYPASS_GND
IGN_OUT
BYPASS_IGN
BTN_TRIG
S14 Key = A
BTN_12V
RL7
RL12
1kΩ
LED2
MIRROR_CLOSE_MOTOR
MIRROR_CLOSE_SW
MIRROR_OPEN_MOTOR
MIRROR_OPEN_SW
RL3
RL4
LED1
+12V_MIRROR_IN
F4 10A
R_BLINK_OUT
F5 10A
L_BLINK_OUT
F6 10A
RL5
RL6
BTN_GND
LOCK_NC
LOCK_COM
LOCK_NO
UNLOCK_NC
UNLOCK_COM
UNLOCK_NO
RL8
LED4
PROTECTION BLOCKS
+12V_ELEC
F2
1.1A
+12V_COILS
BRAKE_12V
R6
30kΩ
D5
3.3V
C4
100nF
R8
10kΩ
R7
PR1
BRAKE_SIG
HB_NEG
R9
4.7kΩ
D6
3.3V
C5
100nF
R11
47kΩ
R10
PR2
HB_SIG
3V3
DOOR_NEG
R12
4.7kΩ
D7
3.3V
C6
100nF
R14
47kΩ
R13
PR3
DOOR_SIG
IGN_LINE
R15
30kΩ
D8
3.3V
C7
100nF
R17
10kΩ
R16
PR4
IGN_LINE_OUT
RELAY BLOCKS
ACC
R18
1kΩ
R21
100kΩ
Q2
2N2222A
D9
1N4007-E3/54
K1
50mH 240Ω
+12V_RAW
ACC_OUT
+12V_COILS
IGN
R19
1kΩ
R22
100kΩ
Q3
2N2222A
D10
1N4007-E3/54
K2
50mH 240Ω
+12V_RAW
IGN_OUT
IGN_LINE
+12V_COILS
START
R20
1kΩ
R23
100kΩ
Q4
2N2222A
D11
1N4007-E3/54
K3
50mH 240Ω
+12V_RAW
START_OUT
+12V_COILS
LOCK
R24
1kΩ
R26
100kΩ
Q5
2N2222A
D12
1N4007-E3/54
K4
50mH 240Ω
LOCK_NC
LOCK_COM
LOCK_NO
+12V_COILS
UNLOCK
R25
1kΩ
R27
100kΩ
Q6
2N2222A
D13
1N4007-E3/54
K5
50mH 240Ω
UNLOCK_NC
UNLOCK_COM
UNLOCK_NO
+12V_COILS
R_BLINK
R28
1kΩ
R30
100kΩ
Q7
2N2222A
D14
1N4007-E3/54
K6
50mH 240Ω
+12V_RAW
R_BLINK_OUT
+12V_COILS
L_BLINK
R29
1kΩ
R31
100kΩ
Q8
2N2222A
D15
1N4007-E3/54
K7
50mH 240Ω
+12V_RAW
L_BLINK_OUT
+12V_COILS
MIRROR_CLOSE
R32
1kΩ
R34
100kΩ
Q9
2N2222A
D16
1N4007-E3/54
K8
50mH 240Ω
MIRROR_CLOSE_SW
MIRROR_CLOSE_MOTOR
+12V_MIRROR_IN
+12V_COILS
MIRROR_OPEN
R33
1kΩ
R35
100kΩ
Q10
2N2222A
D17
1N4007-E3/54
K9
50mH 240Ω
MIRROR_OPEN_SW
MIRROR_OPEN_MOTOR
+12V_MIRROR_IN
+12V_COILS